

|                 |                                                                                    |
|-----------------|------------------------------------------------------------------------------------|
| <b>Ex.No: 3</b> | <b>EXPLORATORY DATA ANALYSIS (EDA) – DATA<br/>CLEANING AND DATA TRANSFORMATION</b> |
| <b>Date:</b>    |                                                                                    |

**AIM:**

**ALGORITHM:**

231401116

## PROGRAM:

```
import pandas as pd
```

```
df = pd.read_csv("indexData.csv")  
print(df.isnull())
```

```
print(df.isnull())
```

```
print(df.isnull().sum())
```

```
df["Open"].fillna(df["Open"].mean(), inplace=True)  
df["High"].fillna(df["High"].mean(), inplace=True)  
df["Low"].fillna(df["Low"].mean(), inplace=True)  
df["Close"].fillna(df["Close"].mean(), inplace=True)  
df["Adj Close"].fillna(df["Adj Close"].mean(), inplace=True)  
df["Volume"].fillna(df["Volume"].mean(), inplace=True)
```

```
print(df)
```

```
print(df.duplicated())
```

```
df = df.drop_duplicates()
```

```
print(df)
```

```
Q1 = df['Open'].quantile(0.25)  
Q3 = df['Open'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR  
upper = Q3 + 1.5 * IQR
```

```
outliers = df[(df['Open'] < lower) | (df['Open'] > upper)]
```

```
print(outliers)
```

```
from sklearn.preprocessing import MinMaxScaler
```

```
scaler = MinMaxScaler()
```

```
df[['Close']] = scaler.fit_transform(df[['Close']])
```

```
print(df[['Close']])
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
print("\nFinal Dataset:")  
print(df)
```

```
print("\nMissing Values After Cleaning:")  
print(df.isnull().sum())
```

OUTPUT:

|        | Index | Date  | Open  | High  | Low   | Close | Adj Close | Volume |
|--------|-------|-------|-------|-------|-------|-------|-----------|--------|
| 0      | False | False | False | False | False | False | False     | False  |
| 1      | False | False | False | False | False | False | False     | False  |
| 2      | False | False | False | False | False | False | False     | False  |
| 3      | False | False | False | False | False | False | False     | False  |
| 4      | False | False | False | False | False | False | False     | False  |
| ...    | ...   | ...   | ...   | ...   | ...   | ...   | ...       | ...    |
| 112452 | False | False | False | False | False | False | False     | False  |
| 112453 | False | False | False | False | False | False | False     | False  |
| 112454 | False | False | False | False | False | False | False     | False  |
| 112455 | False | False | False | False | False | False | False     | False  |
| 112456 | False | False | False | False | False | False | False     | False  |

[112457 rows x 8 columns]

---

|           |       |
|-----------|-------|
| Index     | 0     |
| Date      | 0     |
| Open      | 2204  |
| High      | 2204  |
| Low       | 2204  |
| Close     | 2204  |
| Adj Close | 2204  |
| Volume    | 2204  |
| dtype:    | int64 |

---

|        | Index | Date       | Open        | High        | Low         | Close \     |
|--------|-------|------------|-------------|-------------|-------------|-------------|
| 0      | NYA   | 1965-12-31 | 528.690002  | 528.690002  | 528.690002  | 528.690002  |
| 1      | NYA   | 1966-01-03 | 527.210022  | 527.210022  | 527.210022  | 527.210022  |
| 2      | NYA   | 1966-01-04 | 527.840027  | 527.840027  | 527.840027  | 527.840027  |
| 3      | NYA   | 1966-01-05 | 531.119995  | 531.119995  | 531.119995  | 531.119995  |
| 4      | NYA   | 1966-01-06 | 532.070007  | 532.070007  | 532.070007  | 532.070007  |
| ...    | ...   | ...        | ...         | ...         | ...         | ...         |
| 112452 | N100  | 2021-05-27 | 1241.119995 | 1251.910034 | 1241.119995 | 1247.069946 |
| 112453 | N100  | 2021-05-28 | 1249.469971 | 1259.209961 | 1249.030029 | 1256.599976 |
| 112454 | N100  | 2021-05-31 | 1256.079956 | 1258.880005 | 1248.140015 | 1248.930054 |
| 112455 | N100  | 2021-06-01 | 1254.609985 | 1265.660034 | 1254.609985 | 1258.579956 |
| 112456 | N100  | 2021-06-02 | 1258.489990 | 1263.709961 | 1258.239990 | 1263.619995 |

  

|        | Adj Close   | Volume      |
|--------|-------------|-------------|
| 0      | 528.690002  | 0.0         |
| 1      | 527.210022  | 0.0         |
| 2      | 527.840027  | 0.0         |
| 3      | 531.119995  | 0.0         |
| 4      | 532.070007  | 0.0         |
| ...    | ...         | ...         |
| 112452 | 1247.069946 | 379696400.0 |
| 112453 | 1256.599976 | 160773400.0 |
| 112454 | 1248.930054 | 91173700.0  |
| 112455 | 1258.579956 | 155179900.0 |
| 112456 | 1263.619995 | 148465000.0 |

[112457 rows x 8 columns]

```

0      False
1      False
2      False
3      False
4      False

...
112452  False
112453  False
112454  False
112455  False
112456  False
Length: 112457, dtype: bool

```

|        | Index | Date       | Open        | High        | Low         | Close \     |
|--------|-------|------------|-------------|-------------|-------------|-------------|
| 0      | NYA   | 1965-12-31 | 528.690002  | 528.690002  | 528.690002  | 528.690002  |
| 1      | NYA   | 1966-01-03 | 527.210022  | 527.210022  | 527.210022  | 527.210022  |
| 2      | NYA   | 1966-01-04 | 527.840027  | 527.840027  | 527.840027  | 527.840027  |
| 3      | NYA   | 1966-01-05 | 531.119995  | 531.119995  | 531.119995  | 531.119995  |
| 4      | NYA   | 1966-01-06 | 532.070007  | 532.070007  | 532.070007  | 532.070007  |
| ...    | ...   | ...        | ...         | ...         | ...         | ...         |
| 112452 | N100  | 2021-05-27 | 1241.119995 | 1251.910034 | 1241.119995 | 1247.069946 |
| 112453 | N100  | 2021-05-28 | 1249.469971 | 1259.209961 | 1249.030029 | 1256.599976 |
| 112454 | N100  | 2021-05-31 | 1256.079956 | 1258.880005 | 1248.140015 | 1248.930054 |
| 112455 | N100  | 2021-06-01 | 1254.609985 | 1265.660034 | 1254.609985 | 1258.579956 |
| 112456 | N100  | 2021-06-02 | 1258.489990 | 1263.709961 | 1258.239990 | 1263.619995 |

  

|        | Adj Close   | Volume      |
|--------|-------------|-------------|
| 0      | 528.690002  | 0.0         |
| 1      | 527.210022  | 0.0         |
| 2      | 527.840027  | 0.0         |
| 3      | 531.119995  | 0.0         |
| 4      | 532.070007  | 0.0         |
| ...    | ...         | ...         |
| 112452 | 1247.069946 | 379696400.0 |
| 112453 | 1256.599976 | 160773400.0 |
| 112454 | 1248.930054 | 91173700.0  |
| 112455 | 1258.579956 | 155179900.0 |
| 112456 | 1263.619995 | 148465000.0 |

[112457 rows x 8 columns]

---

|        | Index | Date       | Open        | High        | Low         | Close \     |
|--------|-------|------------|-------------|-------------|-------------|-------------|
| 31952  | HSI   | 2007-07-04 | 22279.83008 | 22307.59961 | 22131.24023 | 22218.55078 |
| 31953  | HSI   | 2007-07-05 | 22272.81055 | 22328.60938 | 22126.97070 | 22252.99023 |
| 31954  | HSI   | 2007-07-06 | 22217.46094 | 22565.42969 | 22081.34961 | 22531.74023 |
| 31955  | HSI   | 2007-07-09 | 22719.99023 | 22818.32031 | 22601.65039 | 22817.42969 |
| 31956  | HSI   | 2007-07-10 | 22833.47070 | 22889.60938 | 22718.13086 | 22885.83984 |
| ...    | ...   | ...        | ...         | ...         | ...         | ...         |
| 106945 | N225  | 2021-05-28 | 28912.53906 | 29194.10938 | 28899.66016 | 29149.41016 |
| 106946 | N225  | 2021-05-31 | 29019.44922 | 29147.71094 | 28791.59961 | 28860.08008 |
| 106947 | N225  | 2021-06-01 | 28998.65039 | 29075.47070 | 28611.25000 | 28814.33984 |
| 106948 | N225  | 2021-06-02 | 28730.81055 | 29003.55078 | 28565.83008 | 28946.14063 |
| 106949 | N225  | 2021-06-03 | 28890.39063 | 29157.16016 | 28879.15039 | 29058.10938 |

|        | Adj Close   | Volume       |
|--------|-------------|--------------|
| 31952  | 22218.55078 | 1.890965e+09 |
| 31953  | 22252.99023 | 2.014865e+09 |
| 31954  | 22531.74023 | 2.258577e+09 |
| 31955  | 22817.42969 | 2.780795e+09 |
| 31956  | 22885.83984 | 3.951114e+09 |
| ...    | ...         | ...          |
| 106945 | 29149.41016 | 8.060000e+07 |
| 106946 | 28860.08008 | 5.380000e+07 |
| 106947 | 28814.33984 | 4.740000e+07 |
| 106948 | 28946.14063 | 7.100000e+07 |
| 106949 | 29058.10938 | 0.000000e+00 |

[6302 rows x 8 columns]

|        | Close    |
|--------|----------|
| 0      | 0.006895 |
| 1      | 0.006873 |
| 2      | 0.006883 |
| 3      | 0.006930 |
| 4      | 0.006944 |
| ...    | ...      |
| 112452 | 0.017349 |
| 112453 | 0.017487 |
| 112454 | 0.017376 |
| 112455 | 0.017516 |
| 112456 | 0.017589 |

[112457 rows x 1 columns]

Final Dataset:

|        | Index | Date       | Open        | High        | Low         | Close \  |
|--------|-------|------------|-------------|-------------|-------------|----------|
| 0      | NYA   | 1965-12-31 | 528.690002  | 528.690002  | 528.690002  | 0.006895 |
| 1      | NYA   | 1966-01-03 | 527.210022  | 527.210022  | 527.210022  | 0.006873 |
| 2      | NYA   | 1966-01-04 | 527.840027  | 527.840027  | 527.840027  | 0.006883 |
| 3      | NYA   | 1966-01-05 | 531.119995  | 531.119995  | 531.119995  | 0.006930 |
| 4      | NYA   | 1966-01-06 | 532.070007  | 532.070007  | 532.070007  | 0.006944 |
| ...    | ...   | ...        | ...         | ...         | ...         | ...      |
| 112452 | N100  | 2021-05-27 | 1241.119995 | 1251.910034 | 1241.119995 | 0.017349 |
| 112453 | N100  | 2021-05-28 | 1249.469971 | 1259.209961 | 1249.030029 | 0.017487 |
| 112454 | N100  | 2021-05-31 | 1256.079956 | 1258.880005 | 1248.140015 | 0.017376 |
| 112455 | N100  | 2021-06-01 | 1254.609985 | 1265.660034 | 1254.609985 | 0.017516 |
| 112456 | N100  | 2021-06-02 | 1258.489990 | 1263.709961 | 1258.239990 | 0.017589 |

|        | Adj Close   | Volume      |
|--------|-------------|-------------|
| 0      | 528.690002  | 0.0         |
| 1      | 527.210022  | 0.0         |
| 2      | 527.840027  | 0.0         |
| 3      | 531.119995  | 0.0         |
| 4      | 532.070007  | 0.0         |
| ...    | ...         | ...         |
| 112452 | 1247.069946 | 379696400.0 |
| 112453 | 1256.599976 | 160773400.0 |
| 112454 | 1248.930054 | 91173700.0  |
| 112455 | 1258.579956 | 155179900.0 |
| 112456 | 1263.619995 | 148465000.0 |

[112457 rows x 8 columns]

Missing Values After Cleaning:

Index 0  
Date 0  
Open 0  
High 0  
Low 0  
Close 0  
Adj Close 0  
Volume 0  
dtype: int64

**RESULT:**